MG2MDCFTA100- ORGANOLEPTIC ASSESSMENT OF FOOD

ANSWER KEY

1. Name the five basic taste qualities.

Answer: Sweet, sour, salty, bitter, and umami.

2. State how sensory evaluation assists in product reformulation.

Answer: Sensory evaluation helps identify changes in taste, aroma, texture, and appearance so that products can be reformulated to improve consumer acceptability. It helps improve food products by identifying consumer preferences and detecting undesirable sensory changes. It guides formulation changes, selecting ingredient levels, and improving prototypes to match consumer expectations.

3. Mention the type of attitude considered unsuitable for panel selection.

Answer: Biased, disinterested, careless, or highly prejudiced attitudes are unsuitable for panel selection.

4. Identify the type of sensory test to which the duo–trio test belongs.

Answer: Difference test (discriminative test).

5. Define Flavor Profile Analysis.

Answer: Flavor Profile Analysis is a descriptive sensory method used to identify and describe the flavour characteristics and intensity of food products. The technique provides a written record of a product's perceptible aroma and flavor components, feeling factors and aftertastes. The panellist characterizes individual aroma and flavor notes in the order perceived and assigns an intensity value using a constant rating scale. Panelists independently examine the product under study, record their impressions of aroma, flavor and aftertastes, then reports to a panel leader in an open discussion. The final flavor profile, upon comparison with an original profile, can show the effect of an ingredient substitution, a processing change packaging, age etc.

6. State whether the triangle test is subjective or objective in nature.

Answer: Objective in nature.

7. State the purpose of conducting a monadic test.

Answer: A monadic test is conducted to evaluate one product independently without comparison to other samples. This test is useful for testing foods that have an after taste or flavour carry over which precludes testing a second sample at the same session

8. Define Numerical Scoring Test.

Answer: Numerical Scoring Test is a sensory method in which panelists assign numerical values to sensory attributes based on quality or intensity.

9. State the role of untrained panellists in the Hedonic Rating Test.

Answer: Untrained panellists express their personal liking or preference for a product in the Hedonic Rating Test.

10. Mention the category under which ranking test is classified.

Answer: Rating test.

11. List the qualifications required for panellists.

Answer: ✓ Sound health without any defects affecting sensory perception; ✓ Average sensitivity; ✓ Capability of independent judgement; ✓ Ability to concentrate, train and learn; ✓ Intellectual curiosity and interest in quality evaluation work; ✓ Willingness to spend time in evaluation and submission to periodic tests; ✓ Freedom from prejudices in respect of a particular food product; and ✓ Food enthusiast having the liking for trying different types of foods.

12. Mention the importance of identifying the problem before conducting a sensory test.

Answer: The first factor to be considered by panel leader is to identify the problem to be looked into i.e., aim of the test should be known. Since sensory organs are simulated simultaneously, for example, aroma and taste, colour and appearance, etc, the objective of testing has to be clear and focussed. The judge has to look for these attributes only. Identifying the problem helps in selecting the appropriate sensory method, panelists, and evaluation procedure for accurate results.

Part B

13. State the effect of temperature on aroma volatility.

Answer: The volatility of aromas is related to the temperature of the food. High temperatures tend to volatilise aromatic compounds, making them quite apparent for Judging; cool or cold temperatures inhibit volatilisation.

- Increase in temperature increases aroma volatility.
- Volatile compounds evaporate more rapidly at higher temperatures.
- Aroma perception becomes stronger with increased release of volatile compounds.
- Extremely high temperature may destroy delicate aroma compounds.
- Low temperature reduces aroma release and sensory perception.

14. Outline the steps involved in conducting a multiple sample difference test.

Answer: In this test, more than one test variable can be evaluated per session but with reduced reliability. Each panellist is served 3-6 samples depending upon the number of test variables. One sample is a known standard. The panellist compares each coded sample with the known standard. One coded sample is a duplicate of the standard. Whatever score the panellist assigns to the blind standard is subtracted from the score he assigns to the test variables. The panellist is not to guess. Direction and degree of difference is also to be judged.

1. Define the objective of the test.
2. Select and prepare samples uniformly.
3. Code the samples randomly.
4. Select suitable panelists.
5. Present samples in random order.
6. Ask panelists to identify differences among samples.
7. Record responses.
8. Analyse and interpret the results statistically.

15. Differentiate between sensitivity–threshold test and dilution test.

Sensitivity–Threshold Test	Dilution Test
Determines the minimum detectable concentration of a stimulus	Determines the extent to which a sample can be diluted before the attribute disappears
Measures sensory sensitivity of panelists	Measures strength/intensity of a sensory attribute
Uses increasing concentrations	Uses progressive dilution
Commonly used for taste and odour threshold studies	Commonly used for flavour or odour evaluation

16. Describe how colour, packaging, and prior information may alter sensory judgement.

- Colour influences expectations about flavour and quality.
- Attractive packaging can create positive impressions and increase acceptability.
- Poor packaging may lead to negative bias.
- Brand name and prior information can affect perception and expectations.
- Psychological factors may influence judgement even before tasting.

17. Discuss the factors causing errors in sensory evaluation.

- Psychological errors such as expectation and suggestion.
- Physiological factors like fatigue, illness, or adaptation.
- Environmental factors including lighting, temperature, and noise.
- Improper sample preparation or serving conditions.
- Lack of training or bias among panelists, Errors in recording and interpretation of results.

Errors Due to Physiological Factors- Health, age, enhancement and suppression of taste sensation

Errors due to Psychological Factors: Anticipatory or expectation errors, Stimulus error (intense colour in fruit juice is considered strong in flavour), Logical error, Halo effect

Part C

18. Define sensory evaluation and state its importance in the food industry.

Answer:

Definition: Sensory evaluation is the scientific method used to measure, analyse, and interpret reactions to the characteristics of food as perceived by the senses of sight, smell, taste, touch, and hearing.

Importance in the Food Industry:

1. Helps in quality control and quality assurance.
2. Assists in product development and reformulation.
3. Determines consumer acceptability and preference.
4. Identifies defects and off-flavours.
5. Helps maintain product consistency.
6. Useful in shelf-life studies.
7. Supports market research and product marketing.
8. Helps compare products with competitors.
9. Assists in selecting raw materials and processing methods.
10. Improves overall customer satisfaction.

19. Explain in detail about paired comparison test.

Answer:

The paired comparison test is a discriminative sensory test used to determine whether a difference exists between two samples based on a specific sensory attribute.

Principle: Two coded samples are presented to the panelists simultaneously. The panelists compare the samples and select the one with more intensity of a specified attribute or identify the preferred sample.

The panel members receive several pairs of samples. These may be different or the same samples in each pair. Samples are always given in code numbers. ➢ Different samples are

given in each pair which differ in the intensity of one characteristic, e.g., sweetness, bitterness or rancidity. In each pair, the sample with more or less intense taste will have to be picked out.

Procedure:

1. Select two samples for comparison.
2. Prepare and serve samples under identical conditions.
3. Randomly code the samples.
4. Present both samples to panelists.
5. Ask panelists to compare based on a specific attribute such as sweetness, saltiness, texture, or preference.
6. Collect responses.
7. Analyse the data statistically.

Advantages: Simple and easy to conduct, Requires less time, Suitable for untrained panelists, Useful for small sensory differences.

Limitations: Only two samples can be compared at a time, Not suitable for evaluating multiple attributes simultaneously, Guessing may affect accuracy.